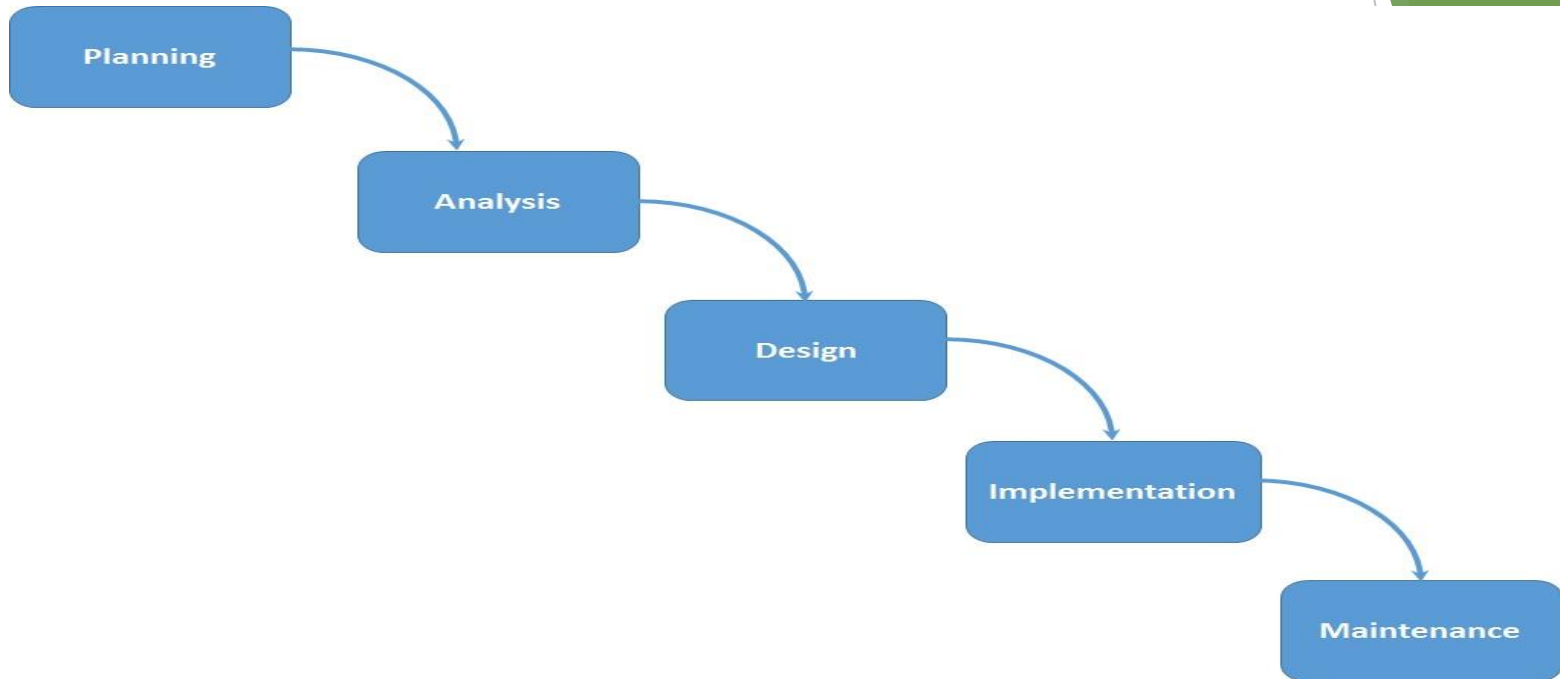


Software development life cycle (SDLC)



- Waterfall process

SDLC

- ▶ Planning the project
 - ▶ Communicate with Stakeholder
 - ▶ Identifying which project need to be automated
 - ▶ Performing a risk analysis associated with the project
 - ▶ Conducting a feasibility study of the project
- ▶ Requirement analysis
 - ▶ Collect information from the project owner through discussion
 - ▶ Gather information from business documents
 - ▶ Interview the existing system's users
 - ▶ Define a high level features of the project
 - ▶ Prepare the detail features and specify the requirements
 - ▶ using different modelling approaches and computer aided software engineering (CASE) tools

SDLC

► Design

- A technical detail of the requirement specifications of the new system is prepared.
- Involves designing the
 - application architecture
 - network architecture
 - user interface
 - Database
 - Navigation architecture and functional design

► Implementation

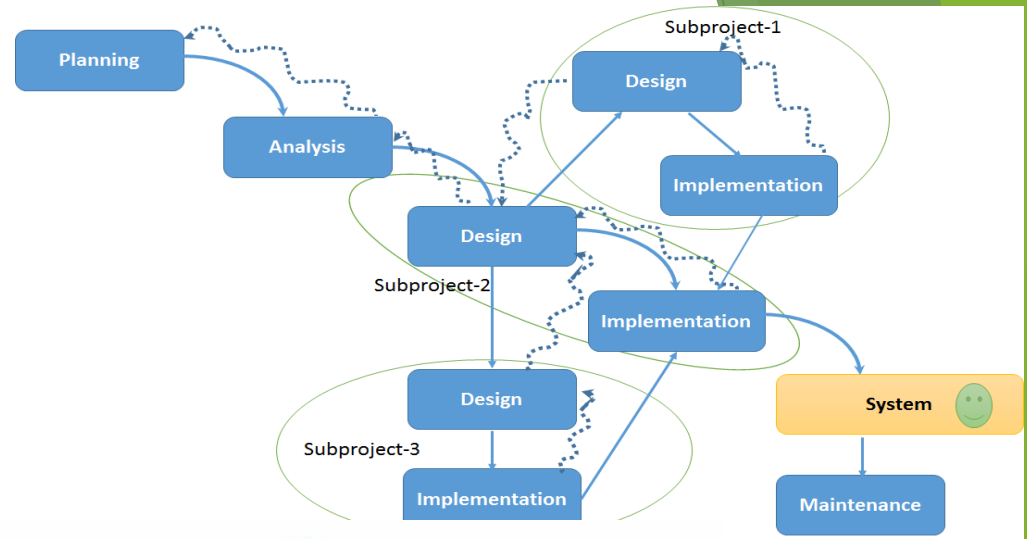
- Design specification is used to write the program for construction and testing
- The purpose of the testing is to confirm that the system satisfies requirements

► Support and maintenance

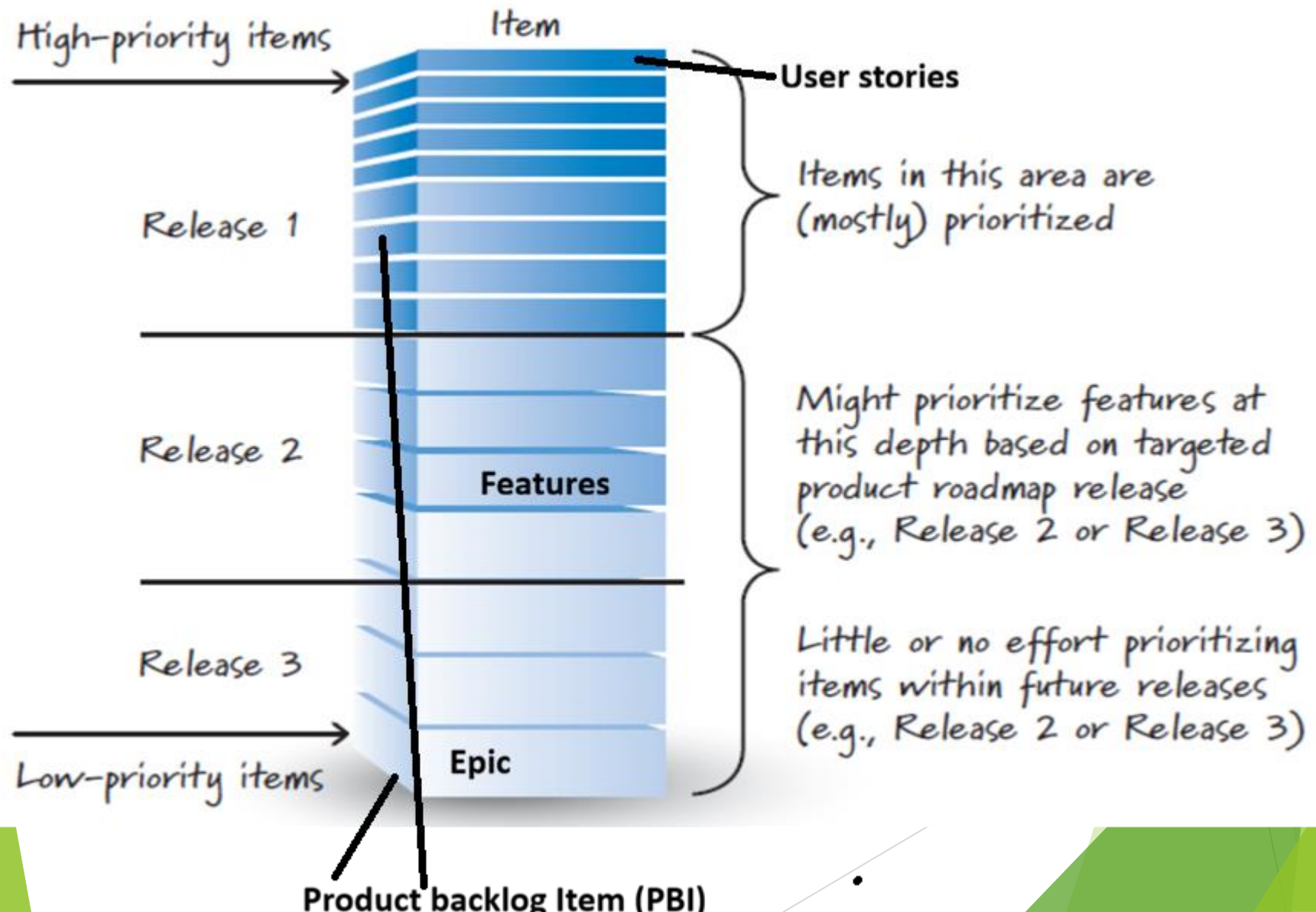
- New systems require monitoring, fixing bugs, modifications or minor adjustments and support while they are in operation.
- New system also may require major changes due to evolving business goal and policies.
- Changes in business policies may happen due to changes in the market conditions or changes in the government policies which organisations need to comply with.
- Some changes may require major upgrades of the system.

Other Software development models

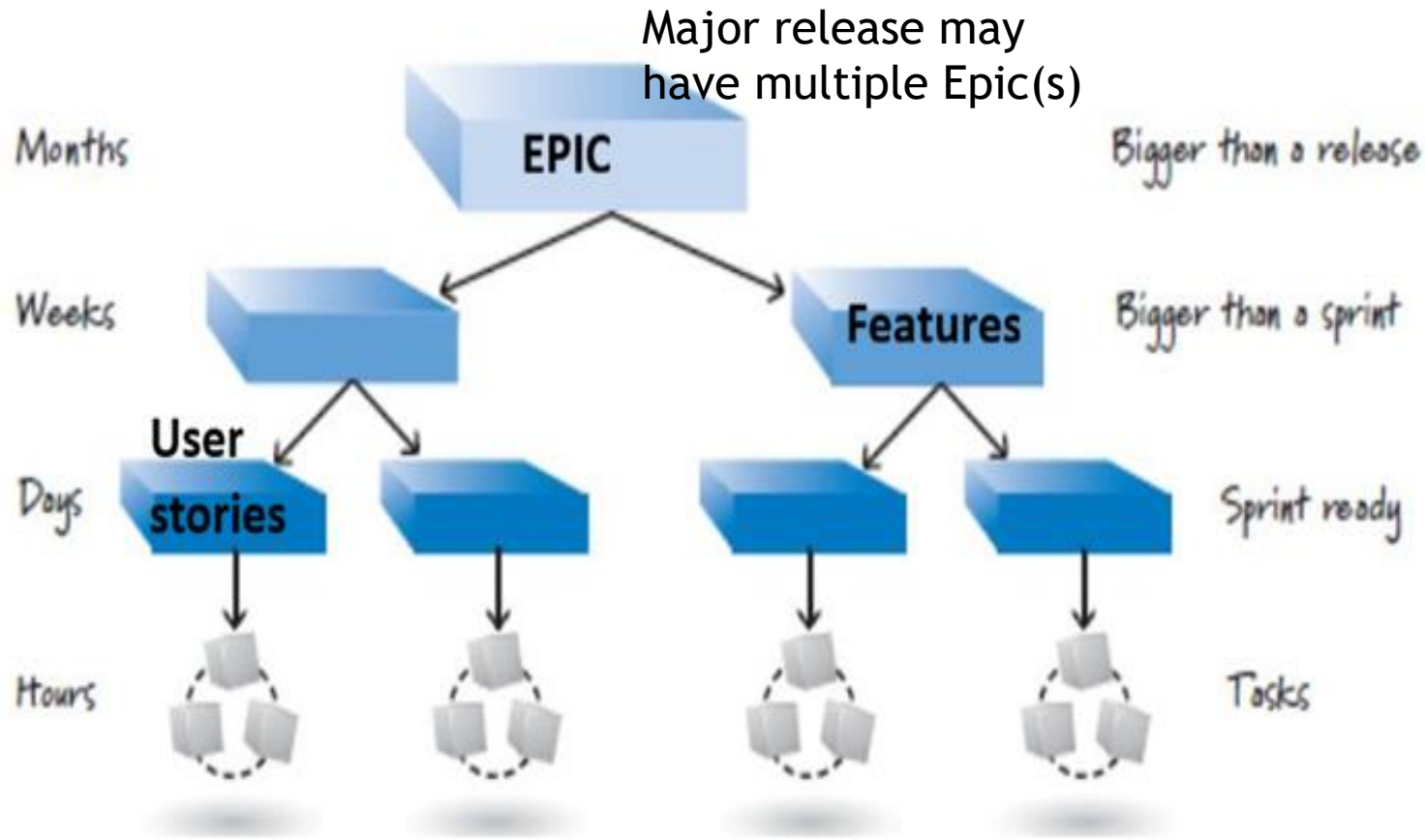
Scrum process model



Product Backlog



Epics, Product Features and user stories



Reference of image: Book-- **Essential Scrum**—Kenneth S. Rubin

Writing your detail requirements in the product backlog: User stories

- ▶ Your epics and features will be decomposed in detail stories according to the artifacts you prepared during backlog grooming stage
- ▶ User story writing template:

Title: Title of user story

As a <User role> I want to <goal> so that <benefit>

Acceptance Criteria: Steps necessary for this story to succeed or be completed

Priority: Any priority

Effort: Unit of effort to complete story

- ▶ **Acceptance criteria: define the boundaries of a user story, and are used to confirm when a story is completed and working as intended.**

User stories as Product Backlog Items (PBIs)

Product Backlog Items (PBIs)/ Description of PBI	Estimate	Priority
As a user I want to create a profile so that I want to see my past purchase history	40 (story points)	9
As a customer I want to be able to add items in the shopping cart so that I can buy multiple items	20	9
As a customer I want to search items so that I can see the detail of product, price and buy the better one.	13	9
As a customer I want to be able to see the items in my cart	40	7
As a customer I want to be able to see the items in my cart	100	5

Estimates provide an indication of the size and therefore the effort and cost of the stories

Writing your detail requirements in the product backlog: User stories

Example of User story-:

As a Customer I want to have a drop-down box in my catalogue so that I can easily see a selection of inventory that I can order the product easily

Acceptance criteria (AC) and other information for this User story-1 :

- Drop-down box needs to be gray

- Drop-down box needs to be 30 pixel in length

- When clicked, drop-down box must show five items in the beginning with the rest below through a scrollbar

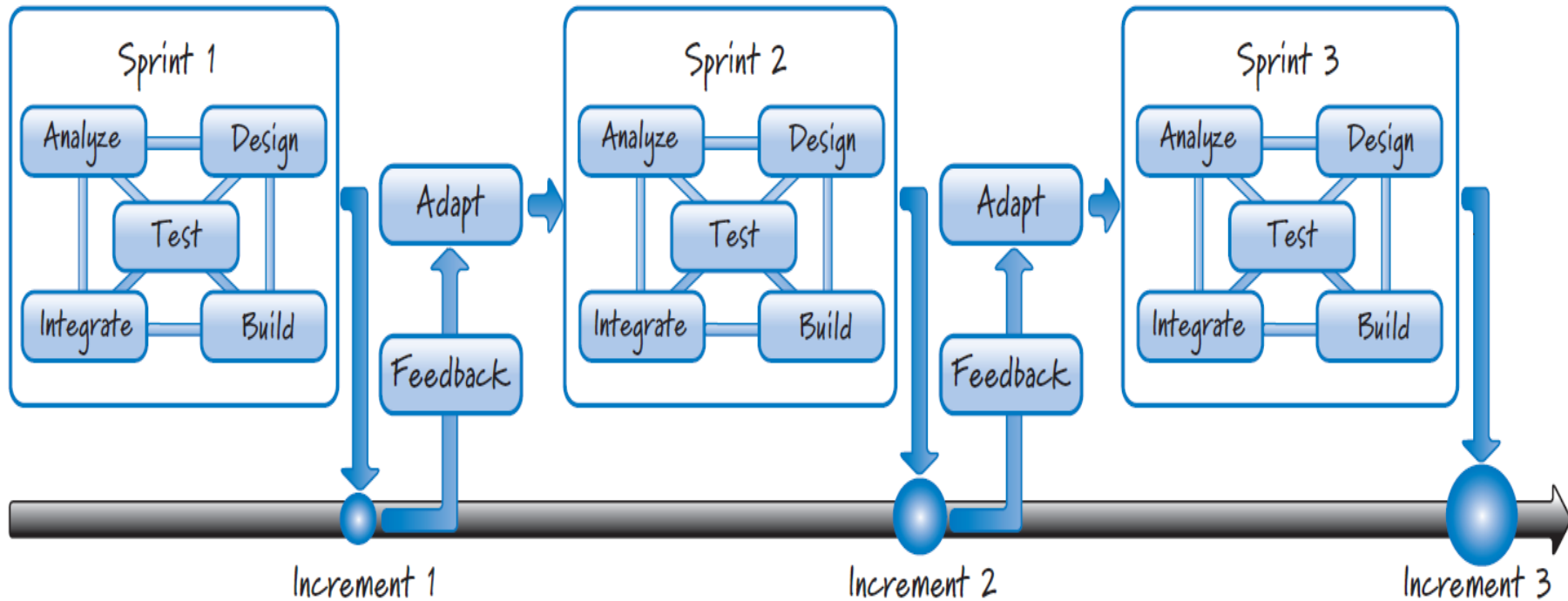
- Items in drop-down box must be in alphabetic order

- Items must also be sorted by size

Priority: 1

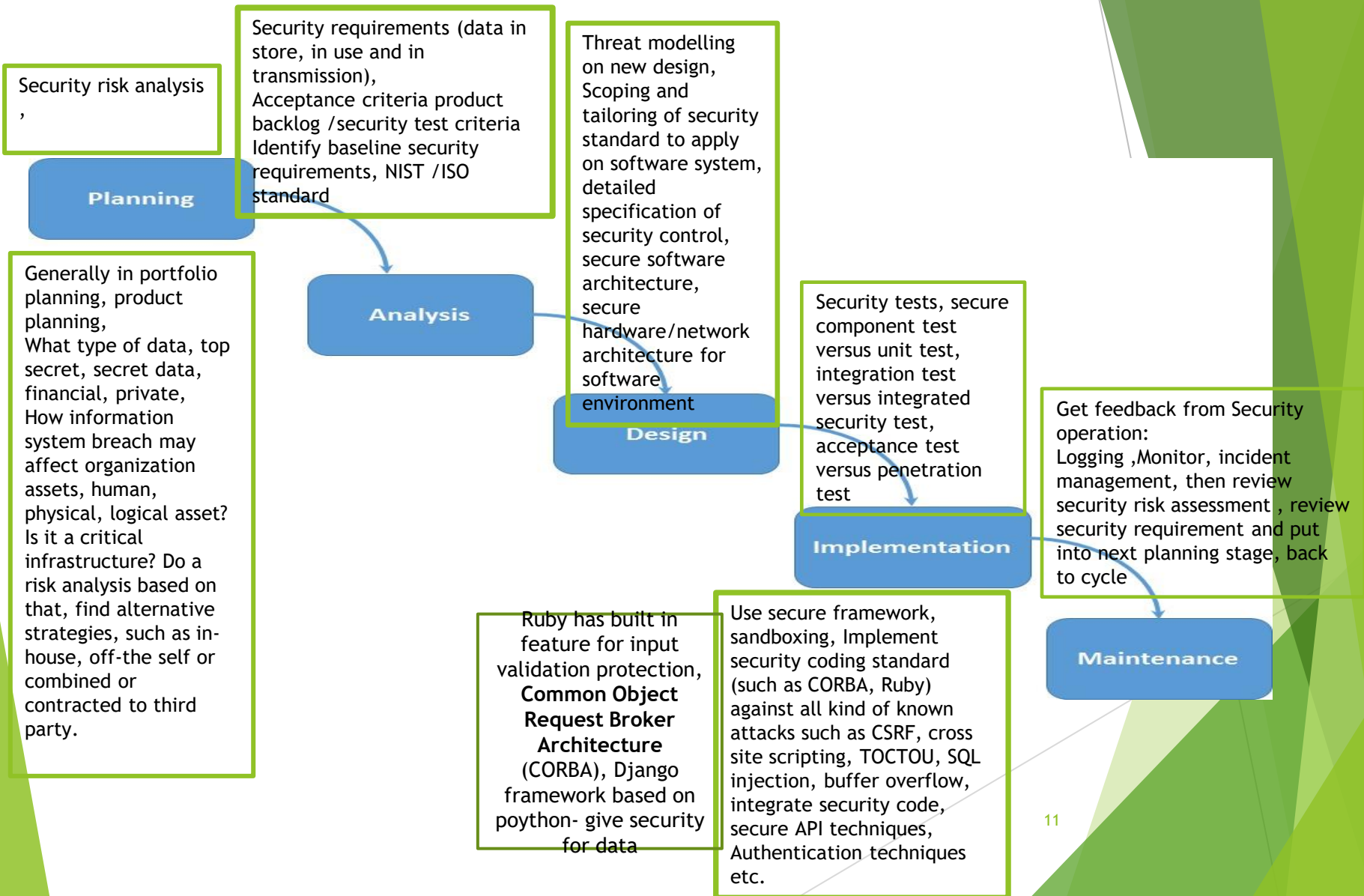
Effort: 4

Sprinting in scrum



- During each sprint we perform all of the activities necessary to create a working product increment (some of the product, not all of it).
- We make some design decisions, create some code based on those decisions, and then test the design and code—all in the same sprint.
- This has the benefit of quickly validating the assumptions that are made when developing product features.
- By doing all of the related work within one sprint, we are able to quickly rework features,

Secure software development life cycle (SSDLC)



References:

Reference:

- ▶ Roger S. Pressman : Software Engineering: A Practitioners Approach. (8th Edition, 2014)
- ▶ Essential Scrum—Kenneth S. Rubin. [Essential Scrum | Agile Alliance](#)
- ▶ The Official (ISC)2 Guide to the CISSP CBK Reference, John Wiley & Sons, Incorporated, 2019
- ▶ Daniel Arrey Arrey, EXPLORING THE INTEGRATION OF SECURITY INTO SOFTWARE DEVELOPMENT LIFE CYCLE (SDLC) METHODOLOGY, PhD Dissertation, Colorado Technical University, USA